



Database Fundamentals Project

Project name: library books

Student name:

رغد إبراهيم الصبحي
امجاد احمد هوساوي
بشرى وصل الله المطرفي
ريما محمد الثبيتي
فوز فواز الشريف
نوره محمد الككبكي



Phase one



Task	Raghad	Amjaad	Noura	Fouz	Bushra	Reema
Business rules				•		•
Chen's Notation	•	•				
UML Notation			•		•	



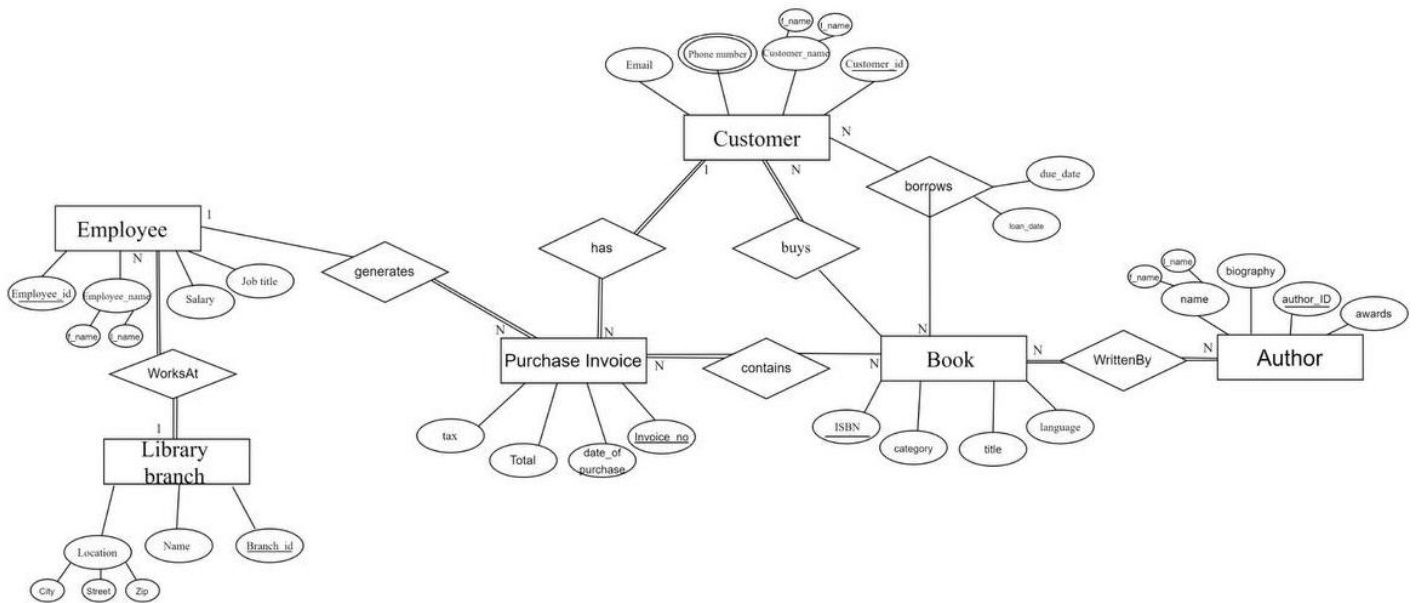
Business rules

Our project is about a library database, containing all the major entities, attributes and the different relationships between them. Here are the business rules for this project:

*The library has many different branches all over the world. Each branch has its own unique id, name and location that contains more information about the exact place like: city, street and zip code. with each branch having multiple employees but each employee works at one single branch. Each employee has his unique id, he also has his first and last name, job title, and salary all stored in the system. Each book has a unique ISBN, a title and language and category . Customers buy books and are issued purchase invoices for the books they buy. Each customer has to buy at least one book and have at least one purchase. Each purchase invoice contains details of one or more books. A book may or may not be bought by a costumer, and it may or may not be contained in a purchase invoice. Each purchase invoice has a unique number, date of purchase, discount amount, and total amount after discount. Each customer has a unique id, a first and last name, and an email address. Customers can have multiple phone numbers. The author has a unique identifier, first name, last name, awards, and personal biography. Each author writes many books, and a book can also have more than one author

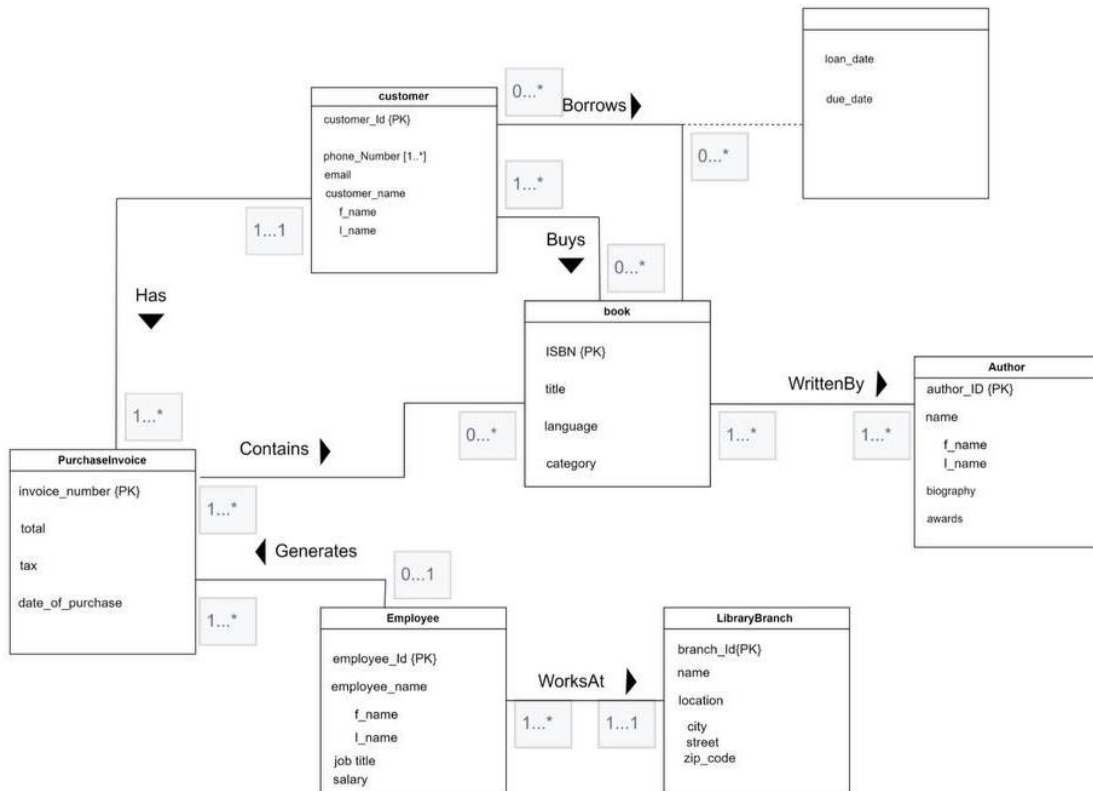


Chen's Notation





UML Notation





Phase two



Task	Raghad	Amjaad	Noura	Fouz	Bushra	Reema
mapping	•	•	•	•	•	•
normalization	•	•	•	•	•	•



Relational schema:

STEP1: Mapping of Regular Entity Types

Employee

<u>Employee_ID</u>	F_name	L_name	salary	Job title
--------------------	--------	--------	--------	-----------

Library branch

City	Street	Zip	Name	<u>Branch_ID</u>
------	--------	-----	------	------------------

Purchase invoice

Tax	Total	Date_of _purchase	<u>Invoice_no</u>
-----	-------	-------------------	-------------------

Customer

Email	C_f_name	C_l_name	<u>Customer_ID</u>
-------	----------	----------	--------------------

Book

<u>ISBN</u>	Category	Title	Language
-------------	----------	-------	----------

Author

Af_name	Al_name	Biography	<u>Author_ID</u>	Awards
---------	---------	-----------	------------------	--------



STEP2: Mapping of Weak Entity Types

No Weak Entities in the ER Model.

STEP3: Mapping of Binary 1:1 Relationship Types

No binary 1:1 relationship in the ER Model.

STEP4: Mapping of Binary 1:M Relationship Types

Purchase Invoice

Tax	Total	Purchase_date	<u>Invoice_ID</u>	Cust_ID
-----	-------	---------------	-------------------	---------

Customer

Email	C_f_name	C_l_name	<u>Customer_ID</u>
-------	----------	----------	--------------------

Purchase Invoice

Tax	Total	Purchase_date	<u>Invoice_ID</u>	Emp_ID
-----	-------	---------------	-------------------	--------

Employee

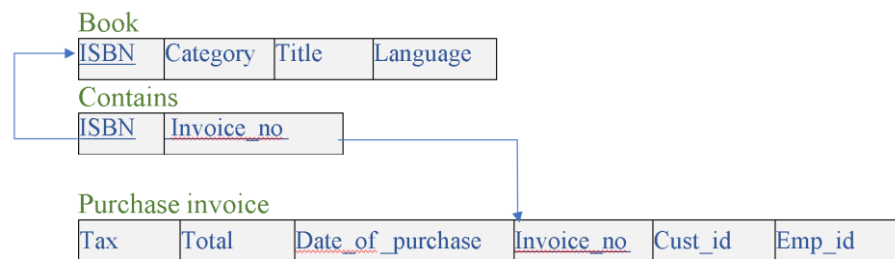
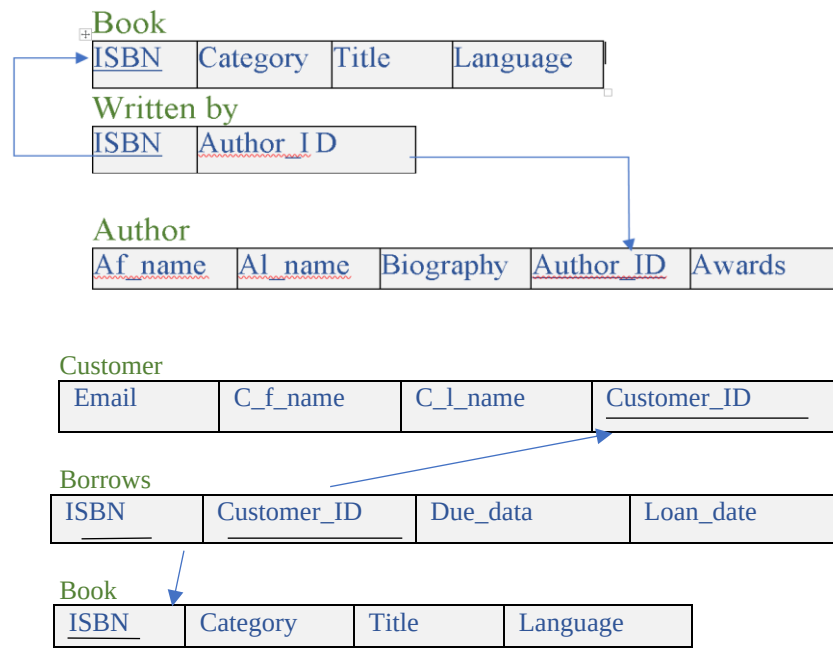
<u>Employee_ID</u>	F_name	L_name	salary	Job title
--------------------	--------	--------	--------	-----------

Employee

<u>Employee_ID</u>	f-name	l-name	Salary	Job title	Bran_ID
--------------------	--------	--------	--------	-----------	---------

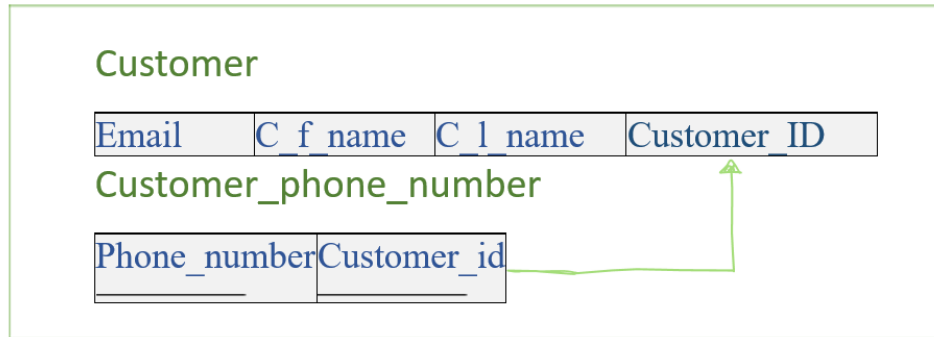
Library branch

City	Street	Zip	Branch_ID	Name
------	--------	-----	-----------	------





STEP6: Mapping of Multivalued Attributes

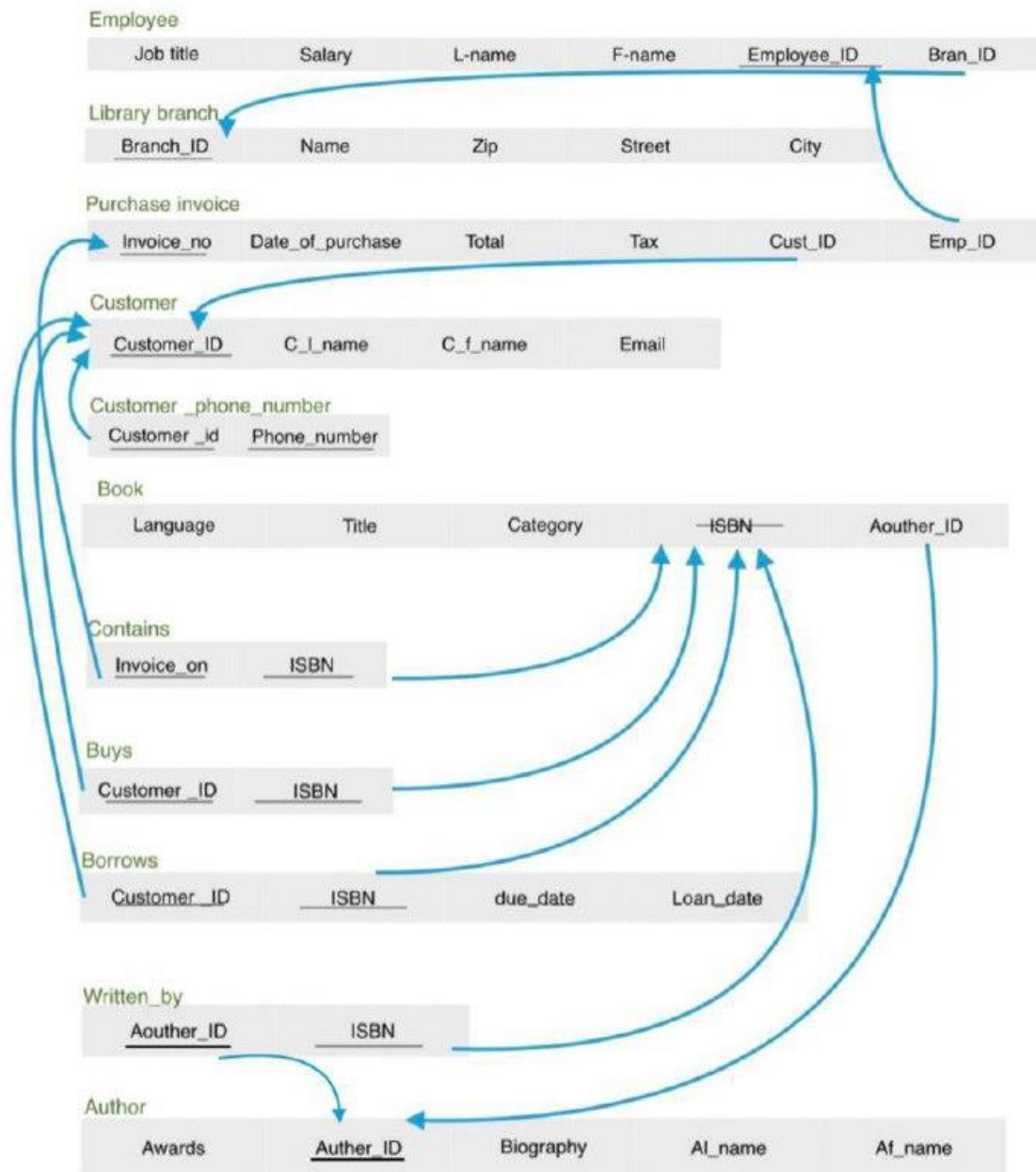


STEP7: Mapping of N-ary Relationship Types

No n-ary relationship in the ER diagram



FINAL MAPPING





NORMALIZATION



First normal (1NF):
No Multivalued Attributes

Employee

<u>Employee_ID</u>	F_name	L_name	salary	Job title
--------------------	--------	--------	--------	-----------

Library branch

City	Street	Zip	Name	<u>Branch_ID</u>
------	--------	-----	------	------------------

Purchase invoice

Tax	Total	Date_of _purchase	<u>Invoice_no</u>
-----	-------	-------------------	-------------------

Customer

Email	C_f_name	C_l_name	<u>Customer_ID</u>
-------	----------	----------	--------------------

Book

<u>ISBN</u>	Category	Title	Language
-------------	----------	-------	----------

Author

Af_name	Al_name	Biography	<u>Author_ID</u>	Awards
---------	---------	-----------	------------------	--------



First normal (2NF):

No Partial Dependency

Employee

<u>Employee_ID</u>	F_name	L_name	salary	Job title
--------------------	--------	--------	--------	-----------

Library branch

City	Street	Zip	Name	<u>Branch_ID</u>
------	--------	-----	------	------------------

Purchase invoice

Tax	Total	Date_of _purchase	<u>Invoice_no</u>
-----	-------	-------------------	-------------------

Customer

Email	C_f_name	C_l_name	<u>Customer_ID</u>
-------	----------	----------	--------------------

Book

<u>ISBN</u>	Category	Title	Language
-------------	----------	-------	----------

Author

Af_name	Al_name	Biography	<u>Author_ID</u>	Awards
---------	---------	-----------	------------------	--------



First normal (3NF):

No transitive dependency

Employee

<u>Employee_ID</u>	F_name	L_name	salary	Job title
--------------------	--------	--------	--------	-----------

Library branch

City	Street	Zip	Name	<u>Branch_ID</u>
------	--------	-----	------	------------------

Purchase invoice

Tax	Total	Date_of _purchase	<u>Invoice_no</u>
-----	-------	-------------------	-------------------

Customer

Email	C_f_name	C_l_name	<u>Customer_ID</u>
-------	----------	----------	--------------------

Book

<u>ISBN</u>	Category	Title	Language
-------------	----------	-------	----------

Author

Af_name	Al_name	Biography	<u>Author_ID</u>	Awards
---------	---------	-----------	------------------	--------